

SEIF Developer Handover & Technical Knowledge Guide

Practical guide for taking ownership of the SEIF Portal, understanding its codebase, authentication, database, business workflows, deployment, and production troubleshooting.

1. Project Architecture

Core: React frontend → REST API → Node/Express backend → MySQL/MariaDB.

Request flow: React Page → API Service → Axios → Express Route → Authentication Middleware → Role Middleware → Controller → Service → Database/S3/Email → Response → React UI

Supporting systems include JWT, role-based authorization, AWS S3, SMTP/Nodemailer, Redis, Socket.IO, cron jobs, Sentry, file uploads, Excel/CSV processing, PDF generation, notifications, reporting, certification/ESSCI, and refurbishment/upgradation.

2. Authentication — Highest Priority

Understand the complete authentication lifecycle first.

- Login and credential validation
- JWT access/refresh tokens
- Role-based authorization
- Partner accounts
- Password reset/resend credentials
- First login and forced password change
- Inactive/suspended accounts
- Protected frontend routes
- Token refresh and logout

Backend: auth.service.js, auth.controller.js, auth.routes.js, auth.middleware.js, role.middleware.js, User.model.js, user.service.js.

Frontend: LoginPage.jsx, auth.api.js, auth.service.js, authSlice.js, useAuth.js, ProtectedRoute.jsx, ForcedPasswordChangeGuard.jsx.

3. Partner Account & Credential Flow

Admin → Organization Management → Create Partner → Partner record → PARTNER user → Temporary password → Credential delivery → Partner login → Forced password change when required.

Important clarification: The Partner creation code appears to generate a temporary password, while the welcome-email sending portion is commented/disabled. Confirm whether production SMTP/welcome-email delivery is active. Also verify the separate resend-credentials flow.

4. Database Architecture

The SQL dump contains a large schema. Start with the major entities rather than memorizing every table.

- users
- partners

- centers
- courses
- batches
- students
- data_uploads
- uploaded_students
- uploaded_centers
- uploaded_batches
- notifications
- audit_logs

Critical relationship: users.partner_id → partners.id. Understand Partner, User, Center, Batch, and Student relationships.

5. Business Modules

Organization Management: Partners, centers, trainers, courses, onboarding, credentials.

Data Upload: CSV/Excel upload, validation, history, approval/rejection, editing, resubmission.

Review & Approval: Partner upload → Admin review → Approve/Reject → Correction → Resubmission.

Certification / ESSCI: Certification uploads, requests, processing, PDFs, approval, archive.

Refurbishment / Upgradation: Requests, packages, rooms, photos, attachments, approvals, notifications.

Notifications: Database notifications, queue/scheduled notifications, Socket.IO, email.

Reports: Identify which API and database queries generate each dashboard/report value.

6. File Handling

Map every module to its storage mechanism: local storage, AWS S3, PDFs, CSV/Excel, certification files, attachments, and generated reports. Ask where original, processed, final, and archived files are stored.

7. Deployment & Production

- Frontend hosting
- Backend hosting
- Database hosting
- Domain/reverse proxy/SSL
- AWS/S3
- Redis
- SMTP/email
- Environment variables
- CI/CD
- Process manager
- Production logs
- Monitoring/Sentry

- Database backup/restore
- Rollback

Critical question: If production goes down and the consultant is unavailable, what should be checked/restarted first?

8. Environment Variables & Secrets

Understand where development, staging, and production secrets are stored. Do not store secrets in ordinary chat. Confirm production does not rely on development fallback secrets/default values.

9. Migrations & Scheduled Jobs

Ask which migration scripts have already run in production and which are historical/obsolete. Never execute migration scripts blindly. Also document cron jobs, schedules, effects, and restart behavior.

10. Questions for the Consultant

Authentication

- Walk through Login → database → JWT → dashboard.
- How do access/refresh tokens, refresh, and logout work?
- Where are tokens stored?
- How are inactive/suspended users handled?
- How does forced password change work?
- How are Partner passwords reset or credentials resent?

Partner

- How exactly is a Partner created?
- When is the users record created?
- How is partner_id linked?
- Where does the temporary password go?
- Is the Partner welcome email working in production?
- What happens if email fails?

Database

- What are the important tables and relationships?
- Which tables are source of truth?
- Which tables must never be manually modified?
- What is the official migration process?
- How is production backup/restore handled?

Upload & Certification

- Explain Upload → Review → Approve → Report.
- What happens when validation fails?
- How does resubmission work?

- Explain the complete Certification/ESSCI lifecycle.
- Where are certification files/PDFs stored?

Deployment & Troubleshooting

- What is the complete production architecture?
- How are frontend/backend changes deployed?
- Where are production logs?
- What is the rollback procedure?
- What are the top 10 known production issues?
- Which code areas are fragile?
- Which old/backup files are unused?
- Which migrations/scripts must not be rerun?

11. Recommended 5-Day Handover Plan

Day 1 — Architecture & Authentication

Project architecture, frontend/backend communication, login, JWT, roles, Partner login, password reset, forced password change.

Day 2 — Database & Organization

users, partners, centers, courses, trainers, batches, students, relationships, Organization Management.

Day 3 — Upload & Review

Upload, validation, Admin review, approval/rejection, correction, resubmission, reporting.

Day 4 — Certification, Refurbishment & Notifications

Complete workflows, files, notifications, and scheduled processing.

Day 5 — Deployment & Troubleshooting

CI/CD, environment, database, AWS/S3, SMTP, Redis, logs, monitoring, backup, rollback, and real bug debugging.

12. Most Valuable Practical Exercise

Ask the consultant to debug a real issue with you from start to finish. For the current Partner login issue, trace: Browser → Network tab → POST /auth/login → request/response → controller → service → User model → users table → bcrypt → JWT → frontend state → protected route.

The goal is to trace a production problem from UI to backend to database and back, not merely memorize files.

13. Personal SELF Knowledge Base

- System Architecture
- Repository Structure
- Local Setup
- Environment Variables
- Database Architecture
- Authentication
- Roles & Permissions
- Partner Management
- Organization Management
- Data Upload Flow
- Review & Approval
- Certification / ESSCI
- Refurbishment / Upgradation
- Notifications
- Reports & Analytics
- File Storage / S3

- Email System
- Scheduled Jobs
- WebSockets
- Deployment Architecture
- CI/CD
- Production Troubleshooting
- Known Bugs
- Technical Debt
- Database Migration
- Backup & Recovery
- Release Process

First priority: Ask the consultant to walk through the complete Partner authentication lifecycle — Partner creation, users record creation, temporary password generation/delivery, login, JWT generation, partner_id linking, forced password change, and Partner API permissions. Verify whether the Partner welcome-email flow is active in production.